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# Seth Peacock

GitHub

740-307-2372

## EDUCATION

**Ohio State University (OSU)** **AU22–AU24**
B.S. in Applied Mathematics, Minor in Philosophy (Summa Cum Laude; College Honors) Cum GPA 3.934/4.0

## RESEARCH EXPERIENCE

**Analyzing self-coherence in oscillations emitted from the ear** **Sept 2024–Present**
*Supervisors: Prof. Christopher Bergevin; Prof. Natasha Mhatre* *York University / The Fields Institute*
Peacock et al., 2025 (preprint on bioRxiv; preparing for submission to PNAS)

- Lead author: wrote sections and edited entire manuscript / integrated collaborators' feedback
- Formulated mathematical framework for our signal processing method
- Engineered a dynamic windowing technique to address time–frequency tradeoff
- Developed documented Python package *phaseco*; integrated all analyses into a unified codebase
- Will be presenting at 2026 Assoc. for Research in Otolaryngology MidWinter Meeting

**Topological data analysis of serotonin time-series data** **Aug 2023–Oct 2024**
*Supervisor: Prof. Janet Best* *OSU*
TDA approach to find differences in serotonin time-series data from depressed vs control mice

- Modified sublevel set filtration (TDA) method to design bespoke algorithm for time-series "process" extraction
- Identified trends in these processes indicating depressed mice lost homeostasis due to measurement electrode

**Fields Undergraduate Summer Research Program** **June 2024–Aug 2024**
*Supervisors: Prof. Christopher Bergevin; Prof. Natasha Mhatre* *The Fields Institute*
Funded summer research program mathematically modeling oscillatory emissions from lizard ears

- Implemented and analyzed stochastic differential equation models of coupled oscillators underlying emissions
- Extended existing model to incorporate interaural coupling between lizard ears

**Optimizing an efficient eigenvector phase retrieval algorithm** **January 2023–May 2023**
*Cycle Undergraduate Research* *OSU*
Designed and implemented methods for efficiently recovering eigenvectors from the absolute value of their entries

## PREPRINTS

**Peacock, S. N. S.**, Vencovsky, V., Whiley, R. E., Mhatre, N., & Bergevin, C. (2025). Spontaneous otocoherence of the active ear. https://doi.org/10.1101/2025.11.14.687084

## SELECTED PROJECTS

***phaseco***
A Python package providing tools for phase autocoherence analysis with dynamic windowing methods

**Peak-picking otoacoustic emissions with machine learning**
A neural network model to identify "peaks" in power spectra of oscillations emitted from the ear

- Designed neural network architecture by adapting and improving models from related applications
- Generated synthetic (labeled) power spectra for supervised learning

**Quantifying uncertainty in college basketball**
Devised Elo-style ranking to derive a bootstrapped C.I. for the minimum "uncertainty" in a game

- 1st Place at OSU's Mathematical Competition in Modeling, November 15th–17th 2024

## PRESENTATIONS

**Assoc. for Research in Otolaryngology MidWinter Meeting (Accepted)** **Feb 2026**
*Otocoherence: Interspecies analysis of phase self-consistency in spontaneous otoacoustic emission*

**Ohio State Honors Project Symposium** **Dec 2024**
*Topological data analysis of depressed mouse serotonin concentrations*

**Fields Undergraduate Summer Research Program Final Presentations** **Aug 2024**
*Modeling and signal processing of spontaneous emissions from lizard ears*

## HONORS AND AWARDS

Conference Travel Award for the 2026 Assoc. for Research in Otolaryngology MidWinter Meeting **2025**
1st Place at OSU's Mathematical Competition in Modeling **2024**
Ohio State Dean's List (All Semesters) **2022-2024**
National Merit Scholar Finalist **2018**

## TEACHING / MENTORING

**Private Tutoring in Mathematics** **Aug 2025–Present**
One-on-one tutoring, primarily in real analysis and proof writing with OSU undergraduates

**Mentoring at Fields Undergraduate Summer Research Program** **June 2025–Aug 2025**
Mentored my supervisor's group in research program I had previously attended *The Fields Institute*

- Proposed accessible entry points based on each student's interests into our research mathematically analyzing and modeling oscillations emitted from ears
- Led regular student meetings for troubleshooting, sharing progress, and developing ideas

**Private Music Lessons** **June 2018–Present**
One-on-one lessons in music theory, guitar, bass, and drums

## SKILLS AND COURSEWORK

**Computational Skills**
Digital signal processing, machine learning, spectral analysis, algorithm design
*Languages:* Python, MATLAB, and Java (completed projects); experience with Julia, Mathematica, JS, C++

**OSU Coursework**
*Applied Mathematics:* Computational Neuroscience, Machine Learning, Statistics, Infectious Disease Dynamics
*Mathematics*: Dynamical Systems, Linear Algebra, ODEs, PDEs, Probability, Real Analysis I & II, Complex Analysis
*Physics*: Classical Mechanics I & II, Electricity and Magnetism, Relativistic Mechanics, Quantum Mechanics

**Self Study**
Signal processing, topological data analysis, linear systems theory, discrete mathematics, circuit theory

## OTHER EXPERIENCE

**Cyclops Studio and Effects:** *Effect Pedal Technician, Studio Engineer* **July 2019–Present**
Repaired, designed, and built electronic musical effect units; engineered and mixed recordings

**Rat Motel:** *Band Manager, Songwriter, Performer* **June 2016–Present**
Managed tours, merchandise, and financial records, developing project management and organizational skills